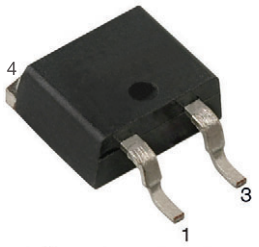
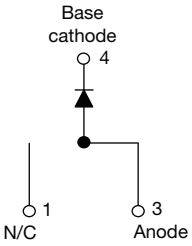
# Hyperfast Rectifier, 20 A FRED Pt® G5




D²PAK 2L (TO-263AB 2L)




## LINKS TO ADDITIONAL RESOURCES

3D Models

Application Notes

| PRIMARY CHARACTERISTICS | |
|---|---|
| $I_{F(AV)}$ | 20 A |
| $V_R$ | 600 V |
| $V_F$ at $I_F$ at 125 °C | 1.23 V |
| $t_{rr}$ (typ.) | 22 ns |
| $T_J$ max. | 175 °C |
| Package | D²PAK 2L (TO-263AB 2L) |
| Circuit configuration | Single |

## FEATURES

RoHS COMPLIANT

HALOGEN FREE

- Best in class forward voltage drop and switching losses trade off
- Optimized for high speed operation
- Minimum creepage and clearance distances are 5.2 mm and 5.4 mm respectively
- Molding Compound Comparative Tracking Index: CTI ≥ 600, provides excellent electrical insulation at relevant working voltages
- 175 °C maximum operating junction temperature
- Polyimide passivation
- Meets MSL level 1, per J-STD-020, LF maximum peak of 245 °C
- Designed and qualified according to JEDEC®-JESD 47
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

## DESCRIPTION / APPLICATIONS

Featuring a unique combination of low conduction and switching losses, this rectifier is the right choice for soft switched and resonant converters, as well as medium frequency hard switching converters. This device is specifically designed to improve efficiency of high speed LLC output rectification stages of EV / HEV battery charging stations and high frequency stages of UPS applications.

## MECHANICAL DATA

**Case:** D²PAK 2L (TO-263AB 2L)
Molding compound meets UL 94 V-0 flammability rating

**Terminals:** matte tin plated leads, solderable per J-STD-002

| ABSOLUTE MAXIMUM RATINGS | | | | |
|---|---|---|---|---|
| PARAMETER | SYMBOL | TEST CONDITIONS | VALUES | UNITS |
| Repetitive peak reverse voltage | $V_{RRM}$ | | 600 | V |
| Average rectified forward current | $I_{F(AV)}$ | $T_C$ = 119 °C, D = 0.50 | 20 | A |
| Repetitive peak forward current | $I_{FRM}$ | $T_C$ = 119 °C, D = 0.50, f = 20 kHz | 40 | |
| Non-repetitive peak surge current | $I_{FSM}$ | $T_C$ = 25 °C, $t_p$ = 10 ms, sine wave | 200 | |
| Operating junction and storage temperature | $T_J$, $T_{Stg}$ | | -55 to +175 | °C |

| ELECTRICAL SPECIFICATIONS ($T_J$ = 25 °C unless otherwise specified) | | | | | | |
|---|---|---|---|---|---|---|
| PARAMETER | SYMBOL | TEST CONDITIONS | MIN. | TYP. | MAX. | UNITS |
| Breakdown voltage, blocking voltage | $V_{BR}$, $V_R$ | $I_R$ = 100 µA | 600 | - | - | V |
| Forward voltage | $V_F$ | $I_F$ = 20 A | - | 1.37 | 1.68 | |
| | | $I_F$ = 20 A, $T_J$ = 125 °C | - | 1.23 | - | |
| Reverse leakage current | $I_R$ | $V_R$ = $V_R$ rated | - | - | 10 | µA |
| | | $T_J$ = 125 °C, $V_R$ = $V_R$ rated | - | - | 500 | |
| Junction capacitance | $C_T$ | $V_R$ = 200 V | - | 25 | - | pF |
| Series inductance | $L_S$ | Measured to lead 5 mm from package body | - | 8 | - | nH |



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| DYNAMIC RECOVERY CHARACTERISTICS ($T_J$ = 25 °C unless otherwise specified) | | | | | | | |
|---|---|---|---|---|---|---|---|
| PARAMETER | SYMBOL | TEST CONDITIONS | | MIN. | TYP. | MAX. | UNITS |
| Reverse recovery time | $t_{rr}$ | $I_F$ = 1.0 A, $dI_F/dt$ = 100 A/µs, $V_R$ = 30 V | | - | 22 | - | ns |
| | | $T_J$ = 25 °C | $I_F$ = 12 A<br>$dI_F/dt$ = 1000 A/µs<br>$V_R$ = 400 V | - | 39 | - | |
| | | $T_J$ = 125 °C | | - | 49 | - | |
| Peak recovery current | $I_{RRM}$ | $T_J$ = 25 °C | | - | 15 | - | A |
| | | $T_J$ = 125 °C | | - | 23 | - | |
| Reverse recovery charge | $Q_{rr}$ | $T_J$ = 25 °C | | - | 290 | - | nC |
| | | $T_J$ = 125 °C | | - | 730 | - | |
| Reverse recovery time | $t_{rr}$ | $T_J$ = 25 °C | $I_F$ = 20 A<br>$dI_F/dt$ = 1000 A/µs<br>$V_R$ = 400 V | - | 47 | - | ns |
| | | $T_J$ = 125 °C | | - | 62 | - | |
| Peak recovery current | $I_{RRM}$ | $T_J$ = 25 °C | | - | 15 | - | A |
| | | $T_J$ = 125 °C | | - | 23 | - | |
| Reverse recovery charge | $Q_{rr}$ | $T_J$ = 25 °C | | - | 400 | - | nC |
| | | $T_J$ = 125 °C | | - | 1000 | - | |

| THERMAL - MECHANICAL SPECIFICATIONS | | | | | | |
|---|---|---|---|---|---|---|
| PARAMETER | SYMBOL | TEST CONDITIONS | MIN. | TYP. | MAX. | UNITS |
| Thermal resistance, junction-to-case | $R_{thJC}$ | | - | - | 1.72 | °C/W |
| Weight | | | - | 2.0 | - | g |
| Maximum junction and storage temperature range | $T_J$, $T_{Stg}$ | | -55 | - | 175 | °C |
| Marking device | | Case style<br>$D^2$PAK 2L (TO-263AB 2L) | E5TH2106S | | | |

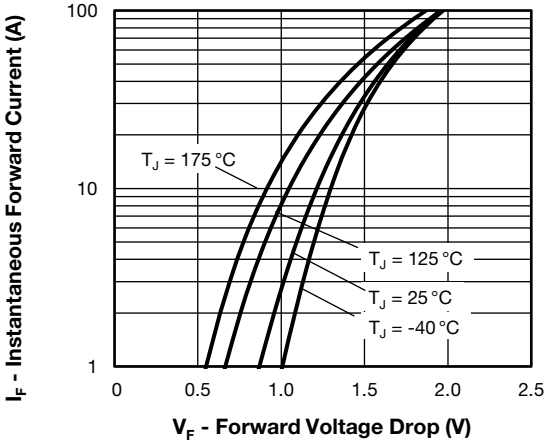



Fig. 1 - Forward Voltage Drop Characteristics

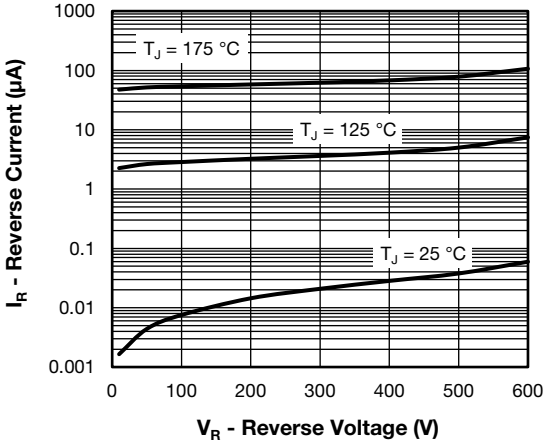



Fig. 2 - Typical Values of Reverse Current vs. Reverse Voltage



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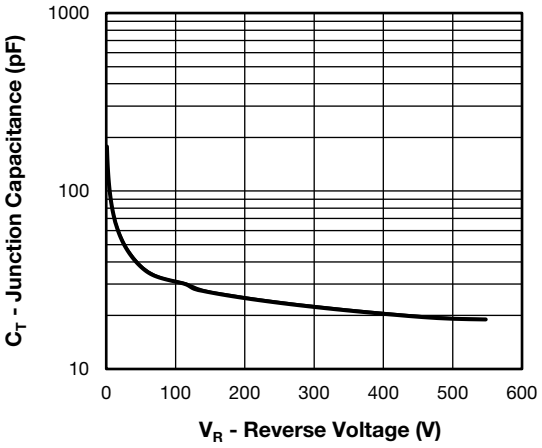



Fig. 3 - Typical Junction Capacitance vs. Reverse Voltage

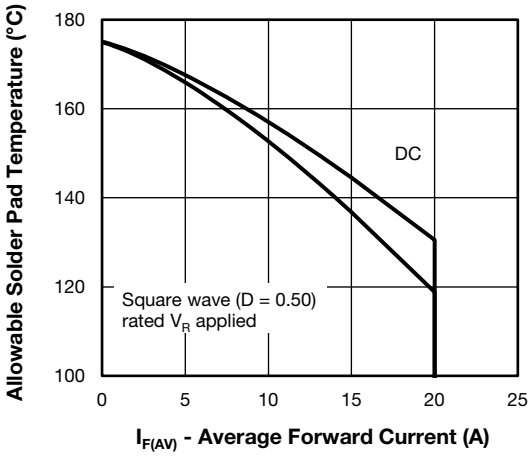



Fig. 4 - Maximum Allowable Case Temperature vs. Average Forward Current

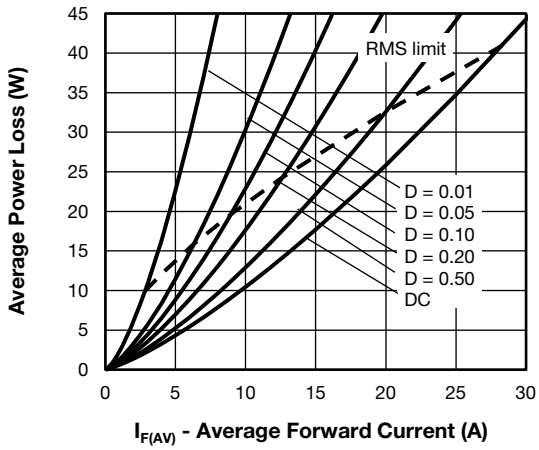



Fig. 5 - Forward Power Loss Characteristics

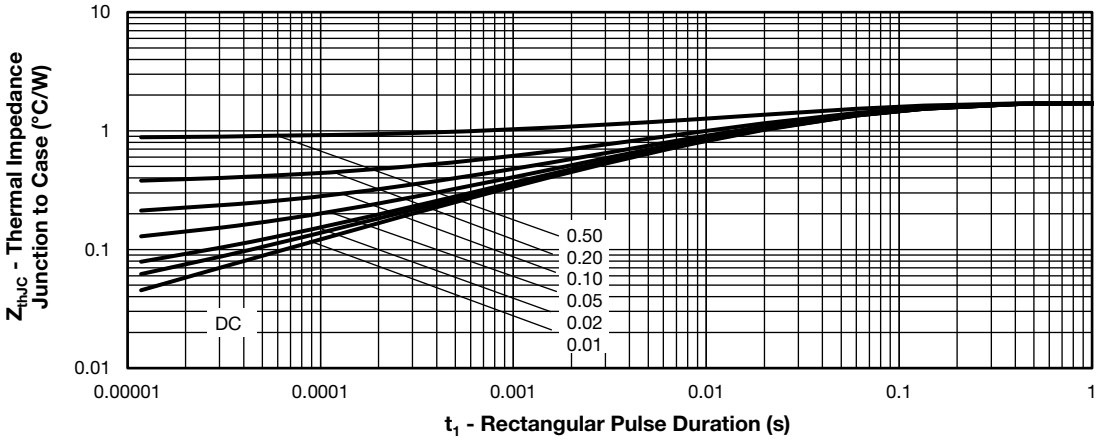



Fig. 6 - Transient Thermal Impedance, Junction to Case


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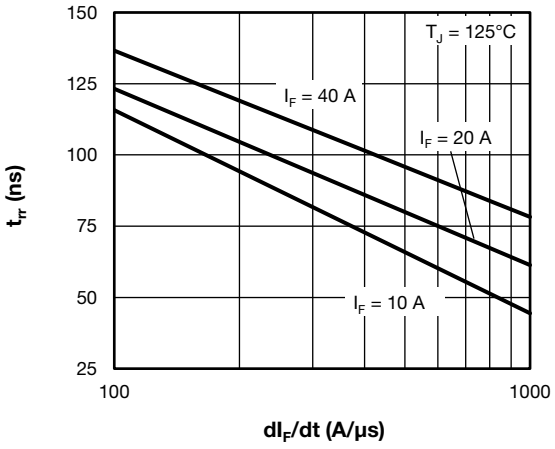



Fig. 7 - Typical Reverse Recovery Time vs. $dI_F/dt$

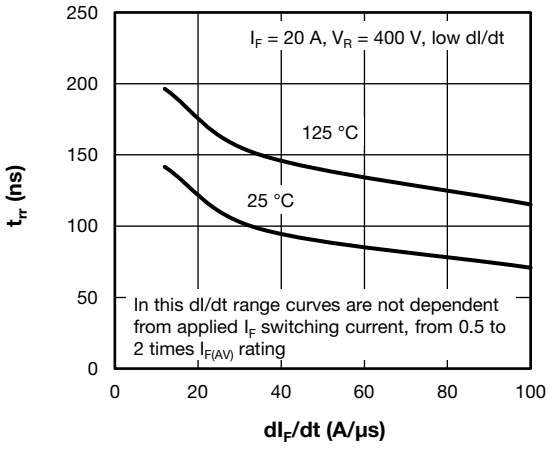



Fig. 10 - Typical Reverse Recovery Time vs. $dI_F/dt$

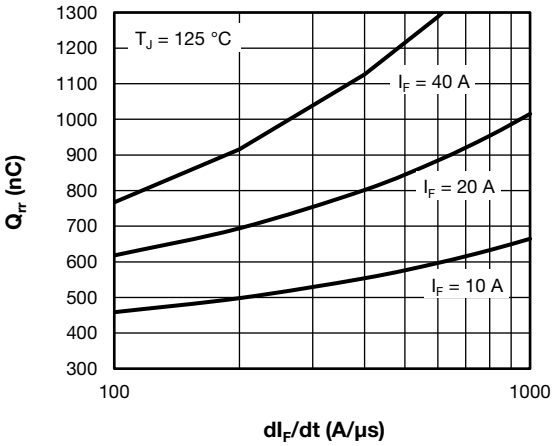



Fig. 8 - Typical Reverse Recovery Charge vs. $dI_F/dt$

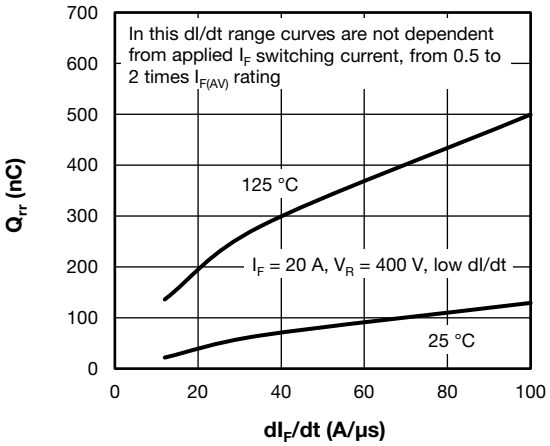



Fig. 11 - Typical Reverse Recovery Charge vs. $dI_F/dt$

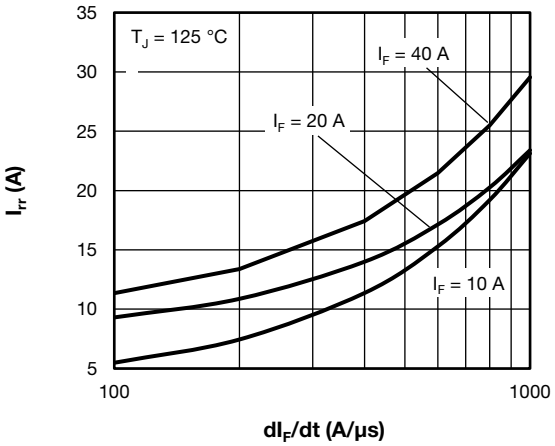



Fig. 9 - Typical Reverse Recovery Current vs. $dI_F/dt$

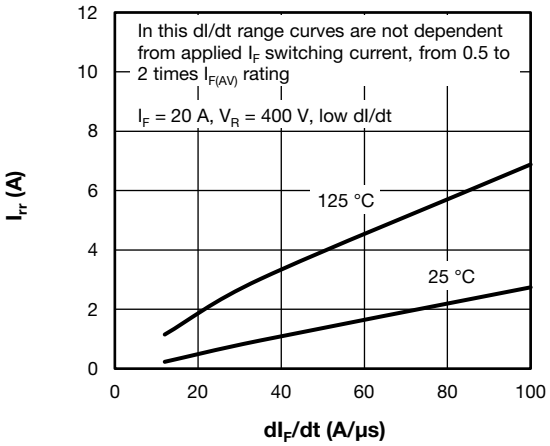



Fig. 12 - Typical Reverse Recovery Current vs. $dI_F/dt$


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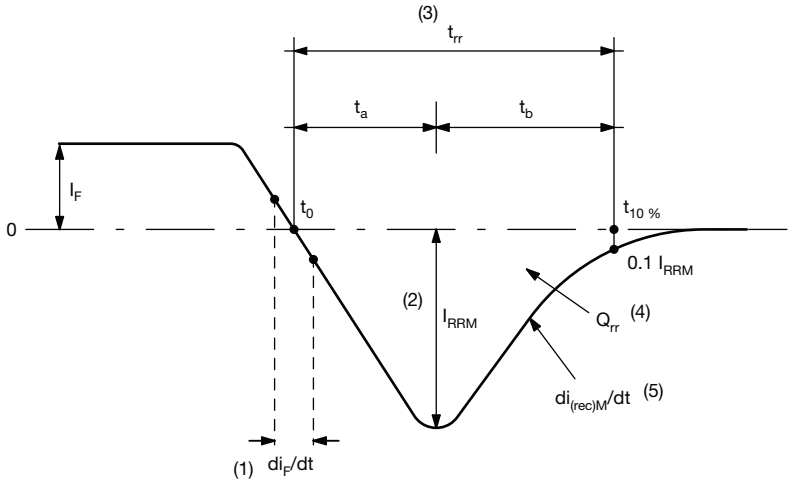



Fig. 13 - Reverse Recovery Waveform and Definitions

**Notes**

(1) $di_F/dt$ - rate of change of current through zero crossing

(2) $I_{RRM}$ - peak reverse recovery current

(3) $t_{rr}$ - reverse recovery time measured from $t_0$, crossing point of negative going $I_F$, to point $t_{10\%}$, 0.1 $I_{RRM}$

(4) $Q_{rr}$ - area under curve defined by $t_0$ and $t_{10\ \%}$

$$Q_{rr} = \int_{t_0}^{t_{10\ \%}} I(t)dt$$

(5) $di_{(rec)}M/dt$ - peak rate of change of current during $t_b$ portion of $t_{rr}$


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## ORDERING INFORMATION TABLE

Device code

| VS- | E | 5 | T | H | 21 | 06 | S2 | L | -M3 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |

1 - Vishay Semiconductors product

2 - E = single diode

3 - 5 = FRED generation 5

4 - Package:
T = $D^2PAK$ (TO-263) package

5 - H = hyperfast recovery

6 - Current rating (21 = 20 A)

7 - Voltage rating (06 = 600 V)

8 - S2 = true 2 pin $D^2PAK$

9 - None = tube (50 pieces)
• L = tape and reel (left oriented, for $D^2PAK$ package)
If needed different orientation/packaging, please contact factory

10 - Environmental digit:
-M3 = halogen-free, RoHS-compliant, and termination lead (Pb)-free

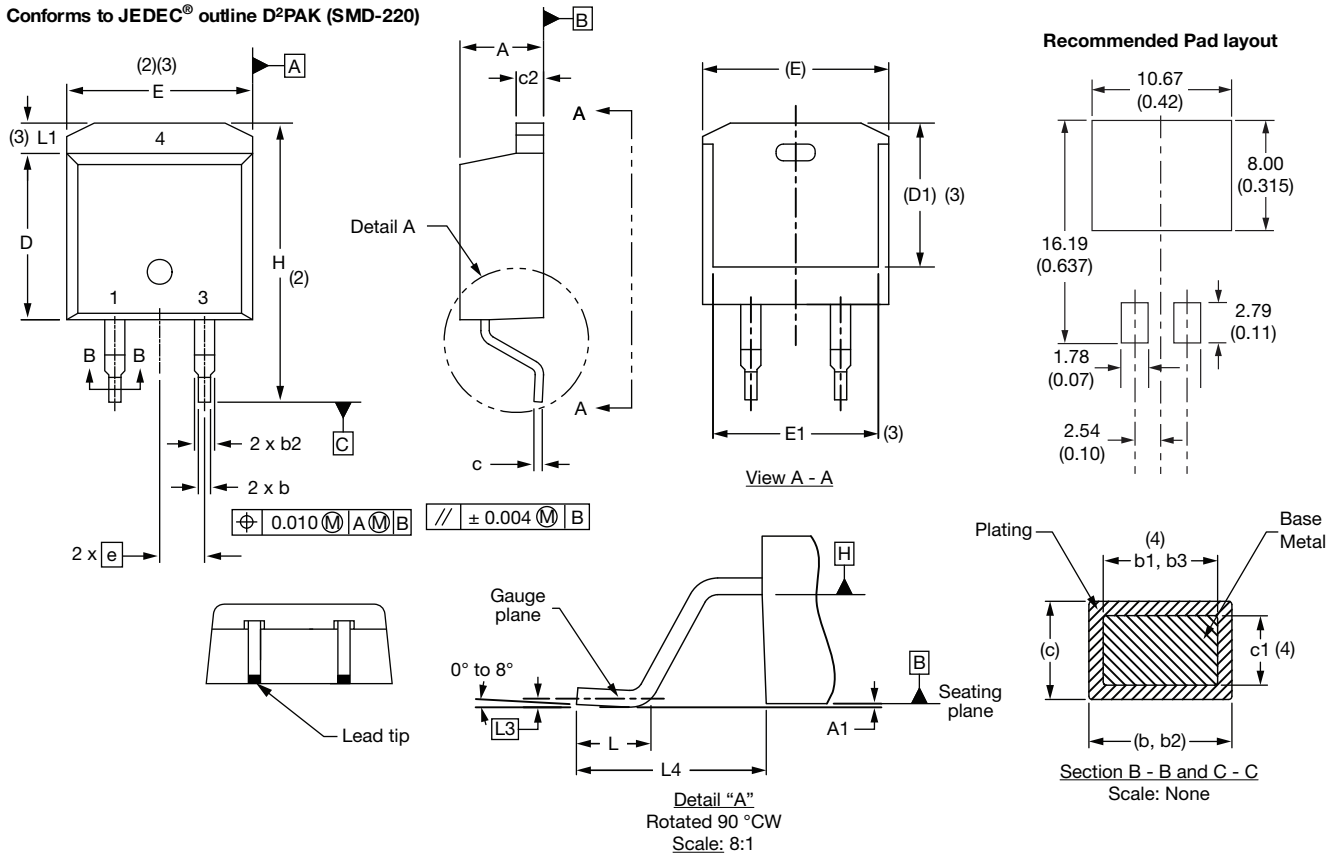
| **ORDERING INFORMATION** (Example) | | |
|---|---|---|
| **PREFERRED P/N** | **BASE QUANTITY** | **PACKAGING DESCRIPTION** |
| VS-E5TH2106S2L-M3 | 800 | 13" diameter reel |

| **LINKS TO RELATED DOCUMENTS** | |
|---|---|
| Dimensions | www.vishay.com/doc?96683 |
| Part marking information | www.vishay.com/doc?96693 |
| Packaging information | www.vishay.com/doc?95032 |


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# $D^2PAK$ 2L (TO-263AB 2L)

**DIMENSIONS** in millimeters and inches




| SYMBOL | MILLIMETERS | | INCHES | | NOTES |
|---|---|---|---|---|---|
| | MIN. | MAX. | MIN. | MAX. | |
| A | 4.06 | 4.83 | 0.160 | 0.190 | |
| A1 | 0.00 | 0.254 | 0.000 | 0.010 | |
| b | 0.51 | 0.99 | 0.020 | 0.039 | |
| b1 | 0.51 | 0.89 | 0.020 | 0.035 | 4 |
| b2 | 1.14 | 1.78 | 0.045 | 0.070 | |
| b3 | 1.14 | 1.73 | 0.045 | 0.068 | 4 |
| c | 0.38 | 0.74 | 0.015 | 0.029 | |
| c1 | 0.38 | 0.58 | 0.015 | 0.023 | 4 |
| c2 | 1.14 | 1.65 | 0.045 | 0.065 | |
| D | 8.51 | 9.65 | 0.335 | 0.380 | 2 |

| SYMBOL | MILLIMETERS | | INCHES | | NOTES |
|---|---|---|---|---|---|
| | MIN. | MAX. | MIN. | MAX. | |
| D1 | 6.86 | 8.00 | 0.270 | 0.315 | 3 |
| E | 9.65 | 10.67 | 0.380 | 0.420 | 2, 3 |
| E1 | 7.90 | 8.80 | 0.311 | 0.346 | 3 |
| e | 2.54 BSC | | 0.100 BSC | | |
| H | 14.61 | 15.88 | 0.575 | 0.625 | |
| L | 1.78 | 2.79 | 0.070 | 0.110 | |
| L1 | - | 1.65 | - | 0.066 | 3 |
| L3 | 0.25 BSC | | 0.010 BSC | | |
| L4 | 4.78 | 5.28 | 0.188 | 0.208 | |

**Notes**

(1) Dimensioning and tolerancing per ASME Y14.5 M-1994

(2) Dimension D and E do not include mold flash. Mold flash shall not exceed 0.127 mm (0.005") per side. These dimensions are measured at the outmost extremes of the plastic body

(3) Thermal pad contour optional within dimension E, L1, D1 and E1

(4) Dimension b1 and c1 apply to base metal only

(5) Datum A and B to be determined at datum plane H

(6) Controlling dimension: inch

(7) Outline conforms to JEDEC® outline TO-263AB



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